

* Phase shift around a loop, as the signal proceeds from input through amplifier feedback network back to input again, completing a loop, is precisely 0° or 360°

* The magnitude of the product of the open loop gain of the amplifier & the magnitude of the feedback factor β is unity.

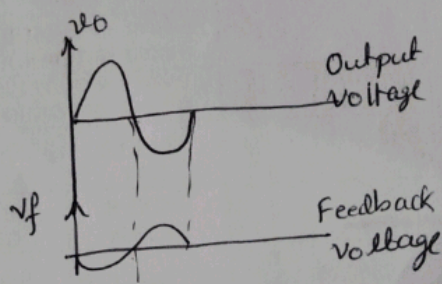
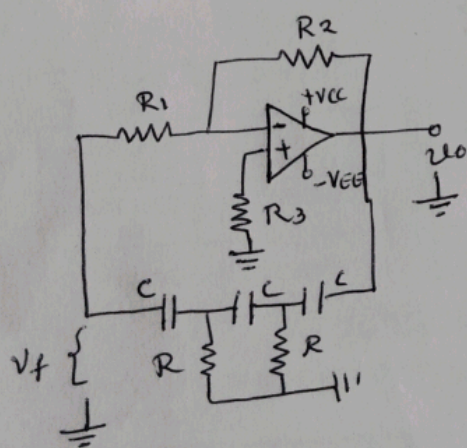
$$|A\beta| = 1$$

Barkhausen Criterion states that $\rightarrow$

* Phase shift oscillators that generate a stable sinusoidal signal at the output.

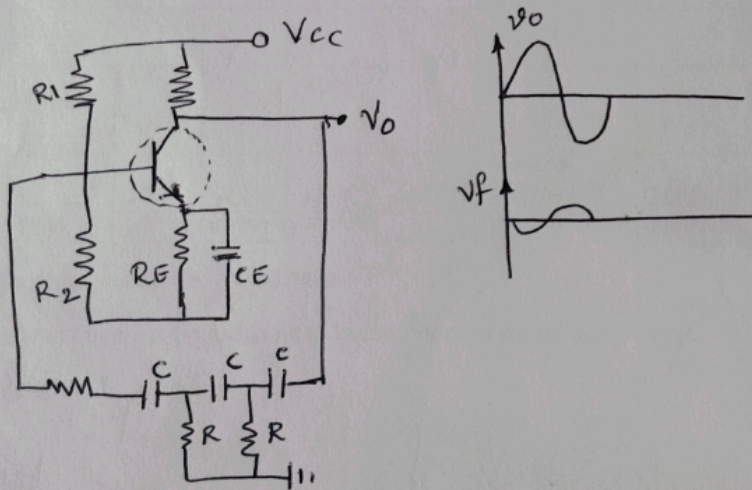
* Basically the circuit has, an amplifier unit (transistor or op-amp) along with a feedback network comprising of resistors & capacitors

* Op-Amp Phase Shift Oscillator



- * Phase shift oscillator, which consists of an inverting amplifier & RC phase-shifting network.
- * The amplifier phase shifts its input by -180° & RC phase lead network phase shifts the amplifier output by a $+180^\circ$ giving a total loop phase shift of zero.
- * The feedback signal is amplified its own input signal to reproduce the output.
- * The circuit is generating its own input signal it is oscillating.
- * The RC phase-lead network consists of 3 equal-value resistors & 3 equal value capacitors
- * The oscillation frequency $f = \frac{1}{2\pi RC \sqrt{6}}$

BJT Phase Shift Oscillator



- * A phase shift oscillator using a single BJT amplifier is shown in fig
- * The amplifier & phase shift network each produce 180° of phase shift.
- * The BJT amplifies the network output &

the network attenuates the amplifier output.

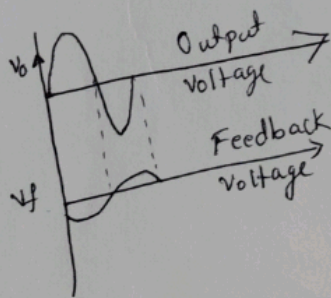
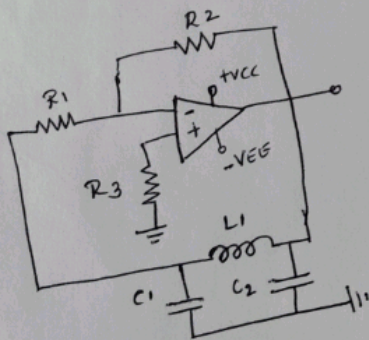
- * The feedback network consists of 3 RC sections each producing 60° phase shift.
- * The output of amplifier is given to feedback network.
- * The output of feedback network drives the amplifier.
- * The frequency of sustained oscillations generated depends on the values of R & C & is given by

$$f = \frac{1}{2\pi\sqrt{6}RC}$$

Op-Amp Colpitts Oscillator

An LC oscillator which uses two capacitive reactances & one inductive reactance in the feedback network i.e. tank circuit is called colpitts oscillator.

- * A colpitts oscillator consists of an inverting amplifier & an LC phase shifting feedback network is shown in fig(1)



- * The LC network acts as a filter that passes the oscillating frequency & blocks all other frequencies.

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* The filter circuit resonates at the oscillating frequency.

For resonance $X_L = X_C$

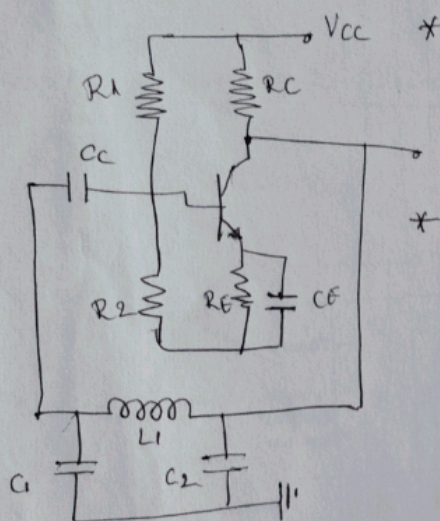
$X_C \rightarrow$ Impedance of the total capacitance in parallel with inductor.

$$\therefore f = \frac{1}{2\pi \sqrt{L, C_T}}$$

$$C_T = \frac{C_1 C_2}{C_1 + C_2}$$

BJT Colpitts Oscillator

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* Colpitts oscillator using an inverting BJT amplifier & an LC feedback network is as shown in fig.

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Hartley Oscillators

* LC oscillator which uses two inductive reactances & one capacitive reactance in its feedback network is called Hartley Oscillator.

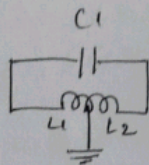
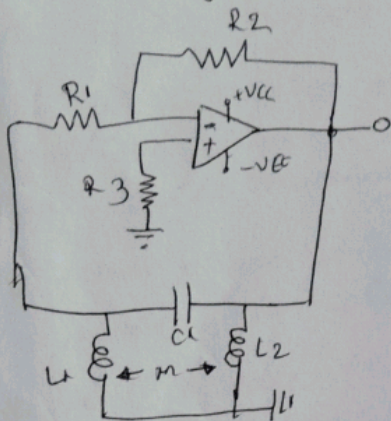
* L_1 & L_2 may be wound on a single core so that there is mutual inductance between the two windings.

$$L_T = L_1 + L_2 + 2M$$

$$\text{Resonance frequency } f = \frac{1}{2\pi \sqrt{C_1 L_T}}$$

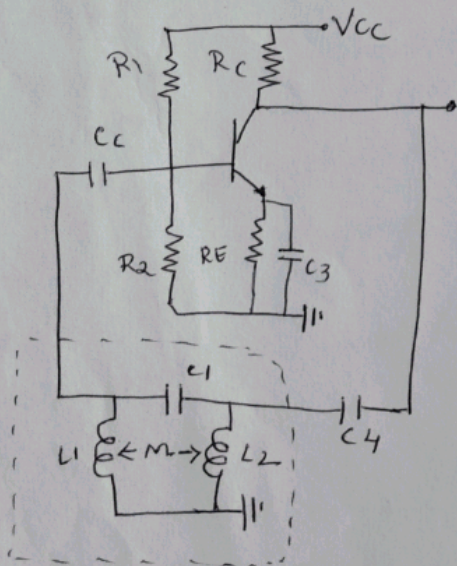
$$\text{Loop gain} = A_{cl(\min)} = L_2/L_1$$

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L_1 & L_2 wound on a single core

BJT Hartley Oscillator



* Common emitter configuration BJT Hartley oscillator is as shown in fig

* The resistances R_1 & R_2 are the biasing resistance.

* The CE amplifier provides a phase shift of 180°

* Coupling capacitors C_c & C_4 are required in order to avoid dc-grounding the transistor base & collector terminals through L_1 & L_2

* L_1 , L_2 and C_1 constitute the phase shift network

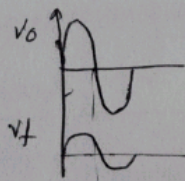
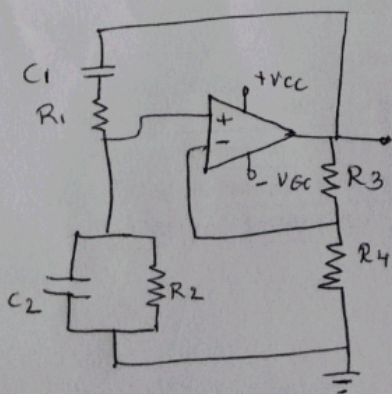
* As emitter is grounded, the base & collector voltages are out of phase by 180°

* So the LC feedback network gives an additional phase shift of 180°

$$\text{Frequency of oscillations } f = \frac{1}{2\pi \sqrt{C_1 L_{eq}}}, \quad L_{eq} = L_1 + L_2$$

Wein Bridge Oscillator

- * In an oscillator, amplifier stage introduces 180° phase shift & feedback network introduces additional 180° phase shift, to obtain total phase shift of 360° .
- * This is the required condition for any oscillator.
- * But Wein bridge oscillator uses a noninverting amplifier & hence does not provide any phase shift during amplifier stage.
- * As total phase shift required is 0° , in Wein bridge no phase shift is necessary through feedback.
- * The Wein bridge is an AC bridge that balances only at a particular supply frequency is as shown in fig.
- * The bridge is made up of all of the resistors & capacitors.
- * The operational amplifier with R_3 & R_4 constitute a noninverting amplifier.
- * The feedback network consists of R_1 , C_1 , C_2 & R_2 .
- * The feedback voltage is in phase with the amplifier output.
- * This voltage is amplified to reproduce the output.
- *
$$f = \frac{1}{2\pi R_1 C_1}$$



Oscillator Amplitude Stabilization.

- * Amplitude stability refers to the ability of the oscillator to maintain a constant amplitude in the output waveform.
- * Amplitude stabilization can indirectly improve frequency stability by ensuring the oscillator operates under consistent condition.
- * To maintain Output Distortion, Frequency stability and output level amplitude stabilization is done.
- * methods for Amplitude stabilization are
 - * Diode stabilization
 - * FET stabilization & many

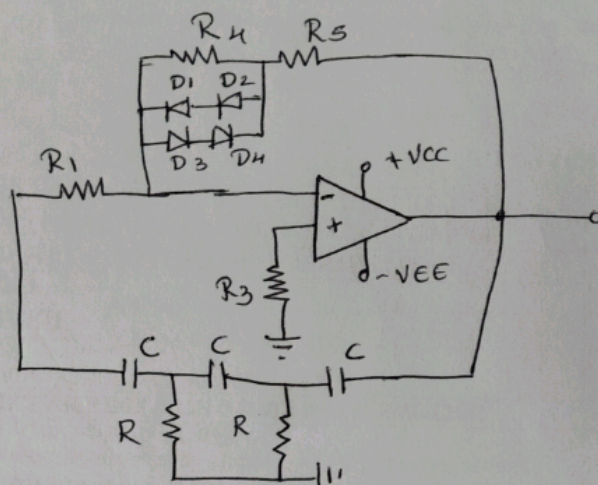
Diode stabilization

- * Diodes can be used to limit the output voltage amplitude
- * When the voltage reaches a threshold, the diodes conduct & output voltage is maintained.

FET stabilization

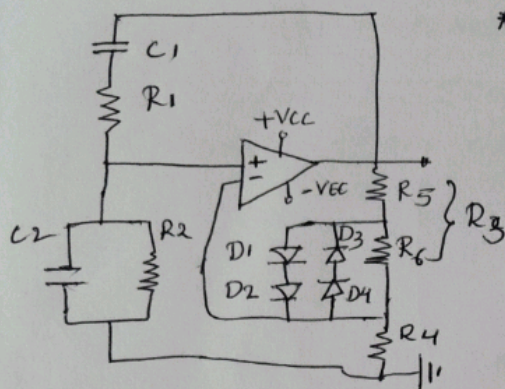
- * FETs can be used as voltage controlled resistors, with their resistance changing based on the gate voltage
- * By using gate voltage, the gain can be controlled.

Diode stabilization circuit for a Phase shift Oscillator



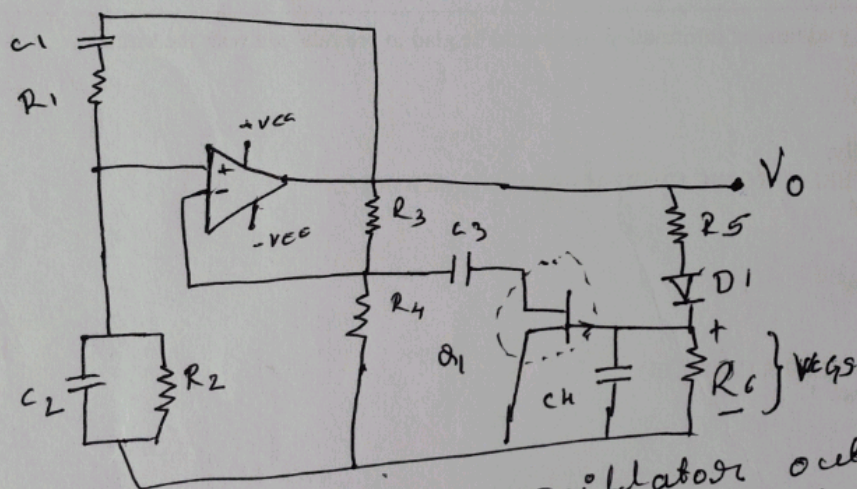
- * A diode stabilization circuit for a phase shift oscillator reduces output amplitude by changing the amplifier's gain when the o/p voltage becomes too high, preventing distortions & ensuring a stable sinusoidal waveform.
- * In above diagram R_2 is bypassed by series-parallel connected diodes.
- * When the output amplitude is low, the diodes do not become forward biased, & no effect on the circuit.
- * Now voltage gain is $A_{CL} = R_2/R_1$
- * when the output amplitude becomes large enough to forward bias either D_1 & D_2 or D_3 & D_4 , R_4 is short circuited & gain will be R_5/R_1 .

Diode stabilization Circuit for Wein Bridge Oscillator



* Resistor R_6 becomes shorted by the diodes when output amplitude exceeds the design level, thus rendering the amplifier gain too low to sustain oscillations

FET stabilization circuit for a Wein Bridge Oscillator



* It stabilizes the oscillator output amplitude by controlling the amplifier gain.

* Capacitor C_3 ensures that Q_1 has no effect on the amplifier dc conditions.

* The output voltage is divided across resistors R_5 & R_6 & rectified by diode D_1 ,

* C_4 smoothes the rectified waveform

- * When the output amplitude is low, V_{gs} is low & this keeps the FET drain-source resistance low.
- * When the output gets larger, V_{gs} is increased, causing R_{DS} to increase. 10
- * The increase in R_{DS} reduces A_{CL} , thus preventing the circuit from oscillating with a high o/p voltage.